
3. Social Media Hyperlink Analytics

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 - Types of Hyperlink Analytics,
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Case Study :

- Do hyperlinks play a role in the popularity of a video posted over YouTube?
- This was the questions that a research team at Centre for Social Technologies (a research center dedicated to understanding social technologies: <http://centreforsocialtech.com/>) set out to explore.
- The research team knew that the answer lies in extracting and visualizing hyperlinks (particularly in-links pointing to a video)
- At the first stage of the quest the research team identified 100 most viewed YouTube videos. Every video posted on YouTube is automatically assigned a unique ID embedded within the URL of the video. For example, this www.youtube.com/watch?v=kffacxfA7G4 is the URL of a video posted by the user “Justin Bieber” having an ID “kffacxfA7G4.”
- Study of **Network of YouTube videos(IDs)** and the domain names pointing to YouTube videos provided the answer

Hyperlinks

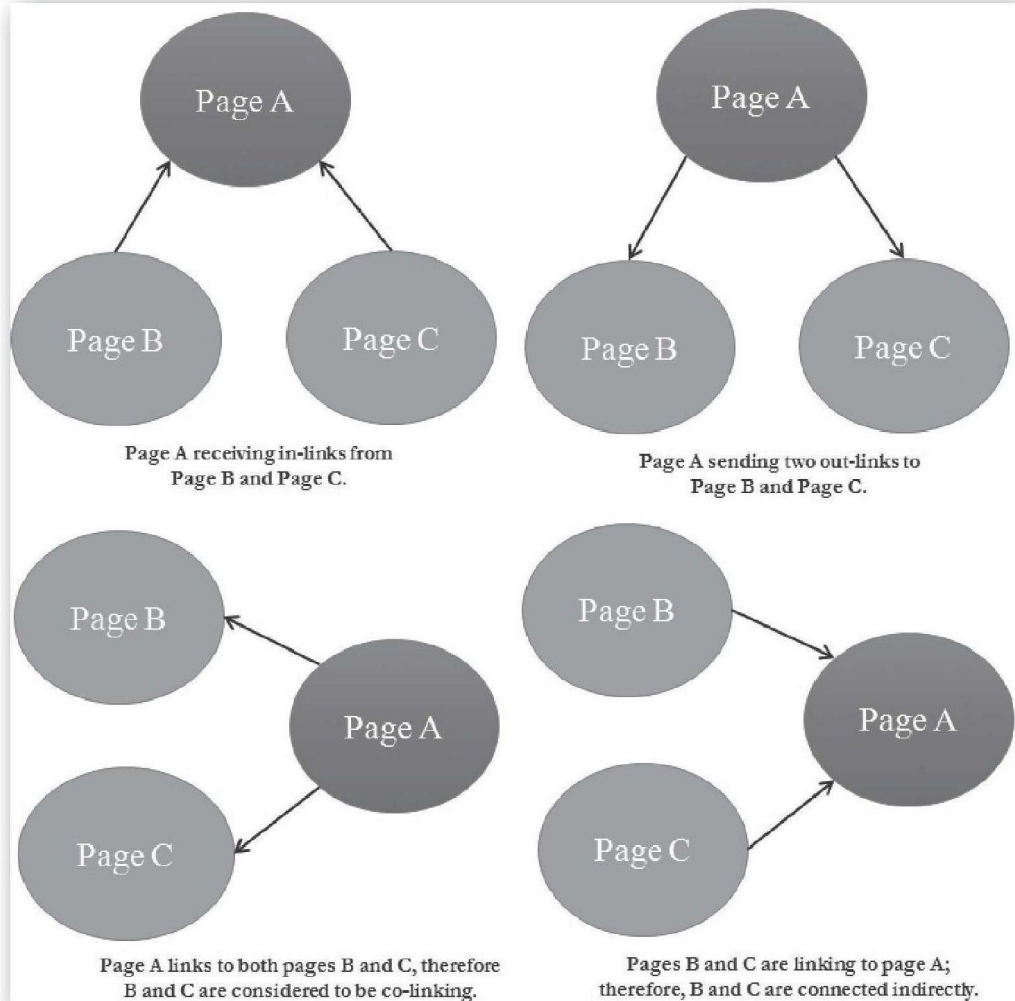
Hyperlinks are **references to web resources** that users can access by clicking on them.

- Text, Image, graphics, icon, email address can be marked as Hyperlink
- Social media hyperlinks are pathways of social media traffic
- They can **link resources within a document (interlinking)** or **among documents (intralinking)**.



Types of Hyperlinks

- IN-LINKS
- OUT-LINKS
- CO-LINKS



Types of Hyperlinks

- **IN-LINKS**

- In-links are the incoming hyperlinks or links directed towards a website or originated in other websites
- In-links are beneficial as they **bring traffic to the website**
- In-links also play an important role in website analytics, **as both the quality and number of in-links can impact the search engine ranking of the website.**

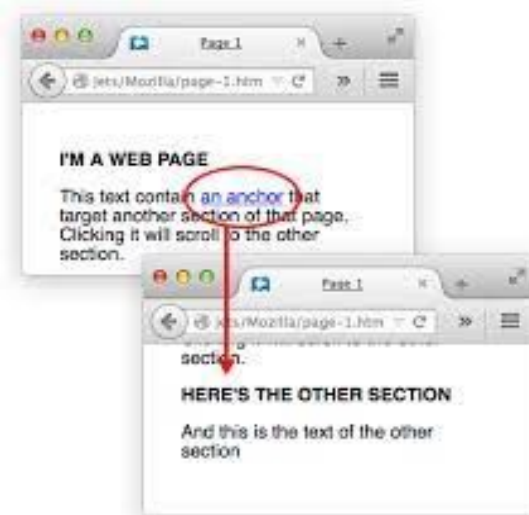
- **OUT-LINKS**

- Out-links are hyperlinks generated out of a website

- **CO-LINKS**

- Co-links have two dimensions.
- First, if two websites receive a link from a third website, they are considered to be connected indirectly. For example, page A links to both pages B and C, therefore B and C are considered to be co-linking, or connected indirectly

- In-links and Out-links :



Hyperlink Analytics

- Hyperlink analytics involves **extracting, analyzing, and interpreting hyperlinks.**
- The **number and quality of hyperlinks to a website** are believed to **reflect its importance or value.**
- Hyperlink analytics can reveal **Internet traffic patterns and sources of incoming or outgoing traffic to and from a website.**
- Hyperlink analysis has been used to study a variety of topics, including **ranking pages, ranking Universities, determining Political networks and measuring business competitiveness.**
- Hyperlink analysis has some limitations, **including that it does not provide insight into the type or amount of traffic flowing among websites within pages.**
- Hyperlinks **within a website between pages** are mostly for navigational purposes and **are given low importance by search engine ranking algorithms.**

Types of Hyperlink Analytics

1. Hyperlink environment analysis
2. Link impact analysis
3. Social media hyperlink analysis.

1. Hyperlink environment analysis

- Hyperlink environment analyses deal with a **particular website or set of websites**.
- Hyperlinks (i.e., out-links, in-links, and co-links) of a website are **extracted and analyzed to identify the sources of Internet traffic**.
- Hyperlinks environment networks can take two forms:
 - **1) co-links networks or**
 - **2) in-links and out-links networks.**

a) Co-Link Networks

- In co-links environment networks, **nodes are websites and links that represent similarity between websites, as measured by co-link counts.**
- With the Webometric Analyst tool, one can construct a **co-link network diagram among a set of websites .**

b) In-Links and Out-Links Networks

- In-links and out-links hyperlink environment networks are constructed based on in-links and out-links from a website or set of websites.
- **Nodes will be websites and links will present in-links and out-links**

2. HyperLink Impact Analysis

- Link impact analysis investigates the impact of a website's URL in terms of **citations or mentions** it receives on the web.
- **Statistics** about **web pages** that mention the **URL** of a given website are **collected and analyzed**.
- It is assumed that a **frequently cited URL is more important**, so measuring the **web impact of URLs** can provide an idea about the importance of a website.

3. Social Media Hyperlink Analytics

- Social media hyperlink analysis involves **extracting and analyzing hyperlinks embedded in social media texts.**
- These hyperlinks can be used to **identify the sources and destinations of social media traffic.**
- A study by Khan et al. found that the **Korean government tended to cite domestic sources and self-cited more, while the US government had a more diverse pattern of out-link destinations.**
- These findings suggest that **social media out-links can carry valuable information and help to explain real-world phenomena and differences in social media use among different cultures.**

Hyperlink Analytics Tool

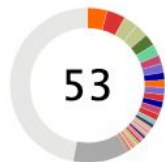
- **Webometric Analyst:** Webometric Analyst is a web impact analysis tool and can conduct variety of analysis on social media platforms including hyperlink network analysis and web mentions: <http://lexiurl.wlv.ac.uk/>
- **VOSON:** VOSON (<http://www.uberlink.com/>) is a hyperlink analytics tools for **constructing and analyzing hyperlink networks**.
- **Open Site Explorer:** Open Site Explorer is a link analysis tool to **research and compare competitor backlinks, identify top pages, view social activity data, and analyze anchor text**: <https://moz.com/researchtools/ose/>
- **Link Diagnosis:** Link Diagnosis (<http://www.linkdiagnosis.com/>) is a free online tool for analyzing and diagnosing links.
- **Advanced Link Manager:** Advanced Link Manager provides a **variety of link analysis capabilities, including the ability to track link-building progress over time, domain quality analysis, backlinks evolution, and website-crawling abilities**: <http://www.advancedlinkmanager.com/>

- **Majestic:** Majestic (<https://majestic.com>) provides a variety of link analysis tools, **including link explorer, backlinks history, and link mapping tools.**
- **Backlink Watch:** Backlink Watch (<http://backlinkwatch.com/>) **is a free tool for checking the quality and quantity of in-links pointing to a website.**

Example Domain

ACTIONS ▾

TRUST FLOW



TOPICAL TRUST FLOW

[... more](#)

35 Arts / Photography

35 Computers / Open Source

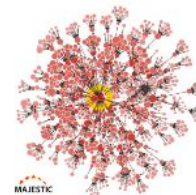
34 Business / Construction and Maintenance

33 Business / Transportation and Logistics

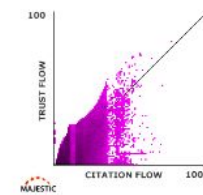
CITATION FLOW



LINK GRAPH

[... more](#)

LINK PROFILE

[... more](#)

EXTERNAL INBOUND LINKS

3,550,997 ▾



FRESH

79,718,888

HISTORIC

663,682,243

REFERRING DOMAINS

442 ^



FRESH

126,617

HISTORIC

1,290,699

REF IPS

856 ^



FRESH

82,888

HISTORIC 479,594

REF SUBNETS

278 ^



FRESH

43,994

HISTORIC 190,234

LIVE

75,658,924

External Inbound Links

DELETED

4,059,964

External Inbound Links

AFTER NOISE REDUCTION ?

119,694 ▾

15,751,588 (-80%)

External Inbound Links



DISTINCT

8,167,359

External Inbound Links

DUPLICATE

67,699,735

External Inbound Links

Trust Flow	Citation Flow
Measures the quality of links pointing to a website or webpage	Measures the quantity of links pointing to a website or webpage
Evaluates the authority and trustworthiness of the linking sources	Reflects the popularity and link equity of a website
Considers the trustworthiness and relevance of the linking sources	Focuses on the number of links, regardless of quality
Based on a "seed set" of trusted websites	Does not require a specific seed set
Provides a numerical value representing link quality	Provides a numerical value representing link quantity
Used to assess the overall quality of a website's backlink profile	Used to assess the overall quantity of a website's backlink profile
Indicates the authority and trustworthiness of a website	Indicates the strength of a website's backlink profile

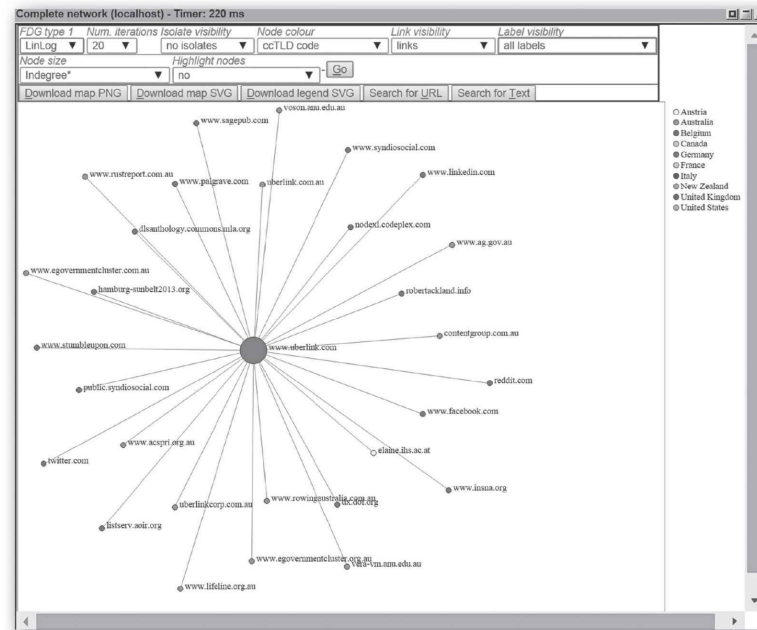
Metric		Analogy	
Citation Flow (CF)		Popularity — many people are talking about you.	
Trust Flow (TF)		Reputation — credible and respected people are talking about you.	
Website	CF	TF	Meaning
Site A	45	40	Strong links, authoritative
Site B	60	10	Many links but low quality → may be spammy
Site C	20	18	Small but trustworthy link profile

Feature	Citation Flow	Trust Flow
Measures	Influence by quantity of links	Authority by quality of links
High Value Means	Many links	Trusted links
Risk	May overrate spam links	Harder to achieve but more meaningful

Hyperlink Analytics Tool

<https://majestic.com/reports/site-explorer>

Demo



Questions

- What are hyperlinks, and why they are important?
- Briefly discuss in-links, out-links, and co-links.
- What is hyperlink analytics and its underlying assumptions?
- What is hyperlink environment analysis?
- What is link impact analysis?
- What is social media hyperlink analysis?